

AWS CLOUD PRACTITIONER HANDBOOK

- ✓ BEGINNER FRIENDLY
- ✓ EXAM FOCUSED
- ✓ REAL WORLD
- ✓ PRACTICAL

YOUR PATH
TO THE CLOUD
STARTS HERE.

FROM CLOUD FUNDAMENTALS
TO **AWS** FOUNDATIONS

CLOUD
MADE SIMPLE.
LEARN. APPLY.
SUCCEED.



COVER ALL
CORE CONCEPTS



EXAM-FOCUSED
CONTENT



REAL-WORLD
EXAMPLES



PRACTICE
SCENARIOS

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BUILD KNOWLEDGE. BUILD SOLUTIONS. BUILD THE FUTURE.



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1. INTRODUCTION TO CLOUD COMPUTING



1 WHAT IS CLOUD COMPUTING?

CLOUD COMPUTING IS THE DELIVERY OF COMPUTING SERVICES OVER THE INTERNET (THE CLOUD). INSTEAD OF OWNING AND MAINTAINING PHYSICAL SERVERS AND INFRASTRUCTURE, YOU ACCESS RESOURCES ON DEMAND FROM A CLOUD PROVIDER LIKE **AWS**.

3 TYPES OF ENVIRONMENTS

ON-PREMISES



All infrastructure is owned and managed by your organization in your data center.

HYBRID CLOUD



Combination of on-premises infrastructure and cloud services working together.

CLOUD ENVIRONMENT



All infrastructure and services are delivered over the internet by AWS.

2 TRADITIONAL IT VS CLOUD COMPUTING

TRADITIONAL IT (ON-PREMISES)	CLOUD COMPUTING (AWS)
HIGH UPFRONT COST	PAY-AS-YOU-GO
HARDWARE PURCHASE & MAINTENANCE	NO HARDWARE TO BUY OR MAINTAIN
LIMITED SCALABILITY	ELASTIC SCALABILITY
LONG PROVISIONING TIME	MINUTES TO PROVISION
YOU MANAGE EVERYTHING	AWS MANAGES THE INFRASTRUCTURE

4 BENEFITS OF CLOUD COMPUTING



COST SAVINGS

Pay only for what you use. No upfront hardware costs.



SCALABILITY

Easily scale up or down based on your needs.



HIGH AVAILABILITY

Built-in redundancy and fault tolerance ensure reliability.



GLOBAL REACH

Deploy applications anywhere in the world.



SECURITY

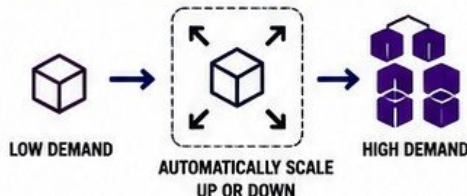
AWS provides robust security and compliance features.



AGILITY

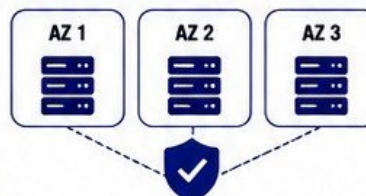
Launch resources quickly and innovate faster.

5 ELASTICITY AND SCALABILITY



CLOUD RESOURCES ADJUST AUTOMATICALLY TO MATCH DEMAND. YOU GET WHAT YOU NEED, WHEN YOU NEED IT.

6 HIGH AVAILABILITY



AWS SPANS MULTIPLE AVAILABILITY ZONES IN A REGION TO ENSURE YOUR APPLICATIONS STAY AVAILABLE EVEN IF ONE ZONE FAILS.

7 PAY-AS-YOU-GO MODEL



YOU PAY ONLY FOR THE RESOURCES YOU USE AND FOR THE TIME YOU USE THEM. NO LONG-TERM COMMITMENTS ARE REQUIRED.



CLOUD COMPUTING HELPS ORGANIZATIONS **REDUCE COSTS**, **INCREASE SPEED**, AND BUILD **RELIABLE**, **SCALABLE**, AND **SECURE** APPLICATIONS.



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2. AWS GLOBAL INFRASTRUCTURE



THE FOUNDATION OF THE AWS CLOUD

1 AWS REGIONS

AWS Regions are geographic areas around the world. Each Region is completely isolated from other Regions to provide the highest levels of availability, security, and compliance.



THERE ARE 30+ AWS REGIONS AROUND THE WORLD.

2 AVAILABILITY ZONES (AZs)

Each Region contains multiple, isolated locations called Availability Zones.

AZs are separated by physical distance to ensure fault tolerance.

REGION



AZ 1



AZ 2



AZ 3

AT LEAST 3 AZs IN EACH REGION

3 EDGE LOCATIONS

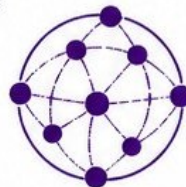
Edge Locations are data centers used by AWS CloudFront to cache content closer to users for lower latency and better performance.



200+ EDGE LOCATIONS WORLDWIDE

4 POINTS OF PRESENCE (POPs)

Points of Presence are network locations that work with Edge Locations to deliver content quickly and reliably.



700+ POPs IN OVER 200 CITIES WORLDWIDE

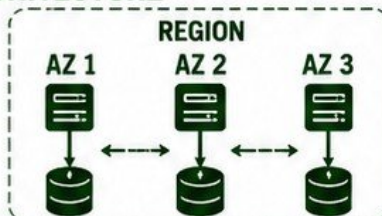
5 CHOOSING A REGION

- ✓ CLOSE TO YOUR USERS FOR LOW LATENCY
- ✓ MEETS COMPLIANCE REQUIREMENTS
- ✓ HAS THE SERVICES YOU NEED
- ✓ CONSIDER PRICING DIFFERENCES
- ✓ PLAN FOR HIGH AVAILABILITY



6 MULTI-AZ ARCHITECTURE

Deploying resources across multiple AZs protects your applications from failures in a single location.



MULTI-AZ = HIGHER AVAILABILITY, FAULT TOLERANCE, AND RESILIENCE

7 GLOBAL VS REGIONAL SERVICES



GLOBAL SERVICES

Work across all Regions. One configuration is global.

- IAM
- Route 53
- CloudFront
- AWS Organizations



REGIONAL SERVICES

Exist in specific Regions. You choose where to deploy.

- EC2
- RDS
- S3
- VPC
- Lambda



AWS GLOBAL INFRASTRUCTURE IS DESIGNED FOR SCALE, PERFORMANCE, AND RELIABILITY. UNDERSTAND IT, AND YOU CAN BUILD ANYWHERE IN THE WORLD WITH CONFIDENCE.

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3. AWS SHARED RESPONSIBILITY MODEL



SECURITY IS A SHARED RESPONSIBILITY

1 AWS RESPONSIBILITY

AWS IS RESPONSIBLE FOR
SECURITY **OF** THE CLOUD.



- ✓ GLOBAL INFRASTRUCTURE
- ✓ REGIONS, AZs, AND EDGE LOCATIONS
- ✓ HARDWARE AND FACILITIES
- ✓ NETWORKING INFRASTRUCTURE
- ✓ HYPERVISORS AND HOSTS
- ✓ MANAGED SERVICES INFRASTRUCTURE

2 CUSTOMER RESPONSIBILITY

YOU ARE RESPONSIBLE FOR
SECURITY **IN** THE CLOUD.



- ✓ YOUR DATA
- ✓ USER ACCESS AND IDENTITY MANAGEMENT
- ✓ OPERATING SYSTEMS
- ✓ NETWORK CONFIGURATION
- ✓ SECURITY GROUPS AND FIREWALLS
- ✓ ENCRYPTION AND KEY MANAGEMENT
- ✓ APPLICATIONS AND CONTENT
- ✓ MONITORING AND LOGGING



SECURITY OF THE CLOUD

AWS is responsible for protecting the infrastructure that runs all AWS services in the Cloud.



SECURITY IN THE CLOUD

You are responsible for protecting your data, applications, and configurations when you use AWS services.

3 RESPONSIBILITY BY SERVICE TYPE

SERVICE TYPE	AWS RESPONSIBILITY (SECURITY OF THE CLOUD)	CUSTOMER RESPONSIBILITY (SECURITY IN THE CLOUD)
INFRASTRUCTURE AS A SERVICE (IaaS) Example: EC2	• GLOBAL INFRASTRUCTURE • HARDWARE • NETWORKING • VIRTUALIZATION	• OPERATING SYSTEM • SECURITY GROUPS • DATA • APPLICATIONS • ACCESS MANAGEMENT
PLATFORM AS A SERVICE (PaaS) Example: Elastic Beanstalk, RDS, Lambda	• GLOBAL INFRASTRUCTURE • HARDWARE • NETWORKING • VIRTUALIZATION • OPERATING SYSTEM • RUNTIME	• DATA • APPLICATIONS • ACCESS MANAGEMENT • NETWORK CONFIGURATION
SOFTWARE AS A SERVICE (SaaS) Example: S3, DynamoDB, CloudFront	• GLOBAL INFRASTRUCTURE • HARDWARE • NETWORKING • VIRTUALIZATION • OPERATING SYSTEM • APPLICATION	• DATA • ACCESS MANAGEMENT • USER PERMISSIONS

4 EC2 VS MANAGED SERVICES

EC2 (IaaS)



- YOU MANAGE: OS, PATCHING, APPLICATIONS, SECURITY GROUPS, FIREWALLS, DATA, BACKUPS, MONITORING.

VS

RDS (Managed Service)



- AWS MANAGES: OS, PATCHING, HARDWARE, BACKUPS, HIGH AVAILABILITY, FAILOVERS.
- YOU MANAGE: DATA, USERS, APPLICATIONS, ACCESS.

5 COMMON RESPONSIBILITY MISTAKES

- ⚠ LEAVING S3 BUCKETS PUBLIC
- ⚠ USING ROOT ACCOUNT FOR DAILY TASKS
- ⚠ WEAK IAM POLICIES (OVER-PERMISSIVE ACCESS)
- ⚠ NOT ENCRYPTING SENSITIVE DATA
- ⚠ NO BACKUPS OR DISASTER RECOVERY PLAN
- ⚠ NOT MONITORING LOGS AND ALERTS
- ⚠ MISCONFIGURED SECURITY GROUPS OR NETWORKS



UNDERSTAND YOUR ROLE. PROTECT YOUR DATA.
SECURE COLLABORATION BUILDS A STRONGER CLOUD.

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4. AWS IDENTITY AND ACCESS MANAGEMENT



MANAGE WHO CAN ACCESS WHAT, SECURELY

1 AWS IAM



AWS IAM

AWS IDENTITY AND ACCESS MANAGEMENT (IAM) IS A WEB SERVICE THAT HELPS YOU SECURELY CONTROL ACCESS TO AWS RESOURCES.

- ✓ CENTRALIZE ACCESS MANAGEMENT
- ✓ CONTROL PERMISSIONS
- ✓ SECURE YOUR AWS RESOURCES
- ✓ COMPLY WITH SECURITY BEST PRACTICES

2 IAM COMPONENTS



USERS

An IAM user represents an individual or application that needs access to AWS.



USER GROUPS

A collection of users. Permissions are assigned to groups.



ROLES

An IAM role is an identity that you can assume to get temporary credentials.



POLICIES

JSON documents that define permissions.



PERMISSIONS

Allows or denies access to AWS actions and resources.

3 HOW IAM WORKS



4 LEAST PRIVILEGE PRINCIPLE



GRANT ONLY THE MINIMUM PERMISSIONS NECESSARY TO PERFORM A TASK.

- ✓ REDUCES SECURITY RISKS
- ✓ PREVENTS ACCIDENTAL CHANGES
- ✓ IMPROVES COMPLIANCE

5 MULTI-FACTOR AUTHENTICATION (MFA)



MFA ADDS AN EXTRA LAYER OF SECURITY TO YOUR AWS ACCOUNTS.

- ✓ SOMETHING YOU KNOW (PASSWORD)
- ✓ SOMETHING YOU HAVE (MFA DEVICE)
- ✓ STRONGER PROTECTION

6 ROOT USER PROTECTION



- ✓ THE ROOT USER HAS FULL ACCESS TO ALL AWS RESOURCES.
- ✓ USE ROOT USER ONLY FOR ACCOUNT SETUP AND BILLING.
- ✓ ENABLE MFA ON THE ROOT USER.
- ✓ CREATE IAM USERS FOR DAILY ACTIVITIES.
- ✓ NEVER SHARE ROOT CREDENTIALS.

7 IAM BEST PRACTICES

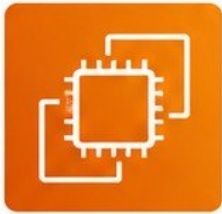


- ✓ USE IAM USERS AND GROUPS FOR PEOPLE.
- ✓ USE ROLES FOR APPLICATIONS AND SERVICES.
- ✓ APPLY LEAST PRIVILEGE.
- ✓ ENABLE MFA.
- ✓ REVIEW PERMISSIONS REGULARLY.
- ✓ MONITOR ACCESS ACTIVITY.



STRONG IDENTITY. CONTROLLED ACCESS. SECURE AWS.
THAT IS AWS IAM.

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5. AMAZON EC2 FUNDAMENTALS

SCALABLE. SECURE. RELIABLE. IN THE CLOUD.



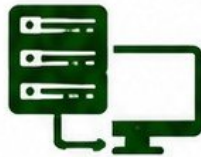
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1 WHAT IS EC2?



Amazon Elastic Compute Cloud (EC2) provides secure, resizable compute capacity in the cloud. You can launch virtual servers, called instances, and run applications in the AWS Cloud.

2 EC2 INSTANCES



EC2 instances are virtual servers in the cloud. You can launch, stop, start, reboot, and terminate instances based on your needs.

3 INSTANCE FAMILIES



Instance families are designed for different use cases such as compute, memory, storage, and accelerated workloads.

4 INSTANCE TYPES



Instance types are sizes within a family that vary by CPU, memory, storage, and network capacity.

5 AMAZON MACHINE IMAGES (AMIs)



AMIs are templates that contain the OS, application server, and applications needed to launch an instance.

6 SECURITY GROUPS



Security groups act as virtual firewalls for your instances. They control inbound and outbound traffic using allow rules (ports and protocols).

7 KEY PAIRS



Key pairs allow you to securely connect to your instances. A public key is stored by AWS, and a private key is kept by you.

8 INSTANCE LIFECYCLE



Launch → Connect → Use → Stop / Start → Terminate. You pay only for what you use.

9 EC2 PRICING OPTIONS



ON-DEMAND INSTANCES

Pay for compute capacity by the second or hour with no long-term commitment. Highest flexibility.



RESERVED INSTANCES

Commit to 1 or 3 years for a significant discount compared to On-Demand. Great for steady workloads.



SPOT INSTANCES

Bid on unused EC2 capacity at a lower price. Ideal for interruptible, fault-tolerant workloads.



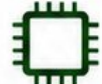
SAVINGS PLANS

Commit to a consistent amount of compute usage (per hour) for 1 or 3 years. Flexible across instance types and regions.



DEDICATED HOSTS

Physical server dedicated for your use. Useful for license compliance and regulatory needs.



DEDICATED INSTANCES

Instance runs on dedicated hardware with no other tenants. Great for compliance and licensing.



EC2 GIVES YOU THE FLEXIBILITY TO LAUNCH VIRTUAL SERVERS, CHOOSE THE RIGHT RESOURCES, AND PAY ONLY FOR WHAT YOU USE.

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6. AWS STORAGE SERVICES



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DURABLE. SCALABLE. SECURE. BUILT FOR THE CLOUD.

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1 AMAZON S3



Amazon Simple Storage Service (S3) is object storage designed to store and retrieve any amount of data from anywhere.

KEY FEATURES

- ✓ Highly durable (11 9s of durability)
- ✓ Virtually unlimited scalability
- ✓ Designed for 99.99% availability
- ✓ Secure, fast, and cost-effective
- ✓ Supports data lakes, analytics, backup, static websites, and more

2 S3 BUCKETS AND OBJECTS



- Buckets are containers for your data in S3.
- Bucket names must be unique across all AWS Regions.
- Objects are the files you store in buckets.
- Each object consists of data, metadata, and a unique key (name).
- Objects can be up to 5 TB in size.

3 S3 STORAGE CLASSES



STANDARD
Frequent access data with low latency and high throughput.



STANDARD-IA
Infrequent access data. Lower cost, minimum 30-day storage.



GLACIER INSTANT RETRIEVAL
Archival data with milliseconds access.



GLACIER FLEXIBLE RETRIEVAL
Archival data with minutes to hours access.



GLACIER DEEP ARCHIVE
Lowest cost storage for long-term archival. Hours to access.

4 AMAZON EBS



Amazon Elastic Block Store (EBS) provides block-level storage volumes for use with EC2 instances.

FEATURES

- ✓ Persistent block storage for EC2
- ✓ High performance and low latency
- ✓ Multiple volume types (gp3, io2, st1, sc1)
- ✓ Data is replicated within an Availability Zone
- ✓ Snapshots for backup and point-in-time recovery

5 AMAZON EFS



Amazon Elastic File System (EFS) provides scalable, managed file storage for use with AWS services and on-premises resources.

FEATURES

- ✓ Shared file storage for multiple EC2 instances
- ✓ Automatically scales as you grow
- ✓ High availability and durability
- ✓ Access via NFS (Network File System)
- ✓ Ideal for lift-and-shift, content management, and home directories

6 INSTANCE STORE



Instance store provides temporary block-level storage for your instance.

FEATURES

- ✓ High performance and very low latency
- ✓ Data is lost when the instance stops, terminates, or hardware fails
- ✓ Ideal for buffer, cache, scratch data, and temporary processing
- ✓ Included with many EC2 instance types at no extra cost

7 STORAGE USE CASES



BACKUPS AND ARCHIVING
Store backups, logs, and archival data in S3 or Glacier for long-term retention.



DATA LAKES AND ANALYTICS
Use S3 to store large datasets for analytics and machine learning workloads.



STATIC WEBSITES AND MEDIA
Host static websites, images, videos, and downloadable content in S3.



APPLICATION STORAGE
Use EBS for application data, EFS for shared files across instances.

8 BACKUP AND LIFECYCLE CONCEPTS



BACKUP
Use S3 versioning, EBS snapshots, and AWS Backup to protect your data and enable point-in-time recovery.



LIFECYCLE MANAGEMENT
Automatically move data between storage classes based on access patterns to reduce costs.



DATA RETENTION POLICIES
Define how long data should be stored and when it should transition or expire.



DURABILITY AND AVAILABILITY
S3 provides 11 9s durability and 99.99% availability. Design for redundancy across AZs and Regions.



CHOOSE THE RIGHT STORAGE FOR THE RIGHT WORKLOAD.
STORE SMART. PROTECT DATA. REDUCE COST.

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7. AWS DATABASE SERVICES

RELIABLE. SCALABLE. SECURE. BUILT FOR EVERY WORKLOAD.

1 AMAZON RDS



Amazon Relational Database Service (RDS) makes it easy to set up, operate, and scale a relational database in the cloud.

KEY FEATURES

- ✓ Supports MySQL, PostgreSQL, MariaDB, Oracle, SQL Server
- ✓ Automated backups
- ✓ Multi-AZ for high availability
- ✓ Read replicas for scaling
- ✓ Easy patching and maintenance

2 AMAZON AURORA



Amazon Aurora is a high-performance relational database built for the cloud with MySQL and PostgreSQL compatibility.

KEY FEATURES

- ✓ Up to 5x more performance than MySQL
- ✓ Built-in high availability
- ✓ Auto-scaling storage
- ✓ Up to 15 read replicas
- ✓ Serverless options available

3 AMAZON DYNAMODB



Amazon DynamoDB is a fully managed NoSQL database that delivers single-digit millisecond performance at any scale.

KEY FEATURES

- ✓ Fully serverless
- ✓ Auto-scaling capacity
- ✓ Built-in security
- ✓ High availability
- ✓ Flexible data model
- ✓ Ideal for key-value and document data

4 AMAZON ELASTICACHE



Amazon ElastiCache provides in-memory caching to improve application performance and reduce database load.

KEY FEATURES

- ✓ Supports Redis and Memcached
- ✓ Microsecond latency
- ✓ High throughput
- ✓ Scales easily
- ✓ Improves application response

5 RELATIONAL VS NoSQL DATABASES

RELATIONAL DATABASES (SQL)



- ✓ Structured data with fixed schema
- ✓ Use SQL for complex queries
- ✓ Strong consistency (ACID)
- ✓ Best for transactions and relationships
- ✓ Examples: RDS, Aurora

NoSQL DATABASES



- ✓ Flexible schema for unstructured data
- ✓ Horizontal scalability
- ✓ High availability and performance
- ✓ Eventual consistency (in most cases)
- ✓ Examples: DynamoDB, ElastiCache

6 MANAGED DATABASE BENEFITS



- ✓ No server provisioning or patching
- ✓ Automatic backups and recovery
- ✓ High availability and fault tolerance
- ✓ Scalable storage and performance
- ✓ Built-in monitoring and security
- ✓ Pay only for what you use
- ✓ Focus on your application, not infrastructure

7 CHOOSING THE APPROPRIATE DATABASE



Use RDS	For traditional relational applications and ACID transactions.
Use Aurora	For high-performance, scalable relational workloads.
Use DynamoDB	For modern applications needing massive scale and flexibility.
Use ElastiCache	For caching, session stores, and real-time leaderboards.



**THE RIGHT DATABASE. THE RIGHT PERFORMANCE.
 THE RIGHT COST. CHOOSE WISELY.**

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8. AWS NETWORKING FUNDAMENTALS

BUILD SECURE. SCALABLE. HIGHLY AVAILABLE NETWORKS IN THE CLOUD.

1 AMAZON VPC



Amazon VPC (Virtual Private Cloud) lets you launch AWS resources in a logically isolated virtual network that you define.

KEY POINTS

- ✓ Isolated virtual network
- ✓ Full control over IP ranges
- ✓ Connect to other AWS services securely
- ✓ Scalable and highly available

2 PUBLIC AND PRIVATE SUBNETS



PUBLIC SUBNET

PRIVATE SUBNET

- Public Subnets have a route to the Internet Gateway. Resources can be accessed from the internet.
- Private Subnets have no direct route to the internet. More secure for internal resources.

3 ROUTE TABLES



- Route tables contain rules that determine where network traffic is directed.
- Each subnet is associated with a route table.
- Local routes are created automatically.

4 INTERNET GATEWAY



- An Internet Gateway allows resources in your VPC to access the internet.
- It is horizontally scaled, redundant, and highly available.

5 NAT GATEWAY



- NAT Gateway allows instances in private subnets to access the internet for updates and downloads.
- Traffic is initiated from the private subnet, not from the internet.

6 SECURITY GROUPS



- Acts as a virtual firewall at the instance level.
- Stateful: allows return traffic automatically.
- Controls inbound and outbound traffic using allow rules.

7 NETWORK ACLs



- Acts as a firewall at the subnet level.
- Stateless: you must allow both inbound and outbound traffic.
- Evaluated before traffic reaches instances.

8 BASIC VPC ARCHITECTURE

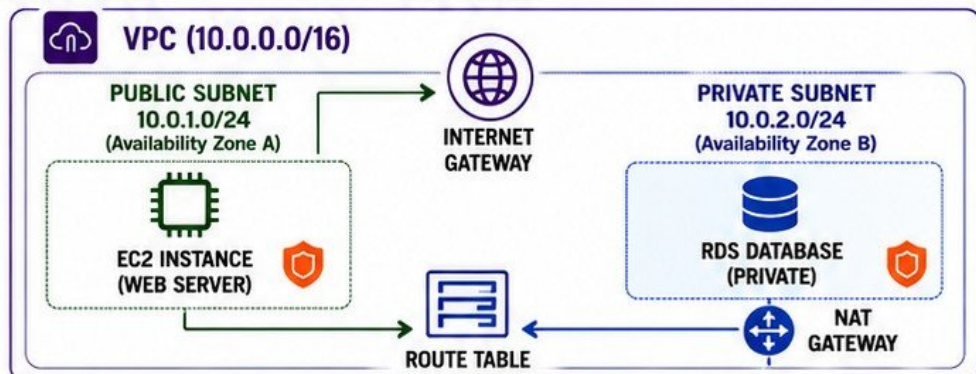


- A VPC contains one or more public and private subnets across Availability Zones.
- Use IGW for internet access and NAT Gateway for private subnets.

9 BASIC VPC ARCHITECTURE EXAMPLE

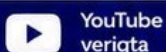
AWS REGION

- VPC
- PUBLIC SUBNET
- PRIVATE SUBNET
- INTERNET GATEWAY
- NAT GATEWAY
- SECURITY GROUP
- ROUTE TABLE



A WELL-DESIGNED NETWORK IS THE FOUNDATION
OF **SECURE**, **SCALABLE**, AND **RELIABLE** CLOUD SOLUTIONS.

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9. ELASTIC LOAD BALANCING AND AUTO SCALING



DISTRIBUTE TRAFFIC. SCALE AUTOMATICALLY. STAY HIGHLY AVAILABLE.

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1 ELASTIC LOAD BALANCING (ELB)



Elastic Load Balancing automatically distributes incoming application traffic across multiple targets (EC2 instances, containers, or IPs) to improve availability, fault tolerance, and scalability.

KEY BENEFITS

- ✓ Distributes traffic across healthy targets
- ✓ Improves application availability and fault tolerance
- ✓ Integrates with Auto Scaling for elasticity
- ✓ Handles millions of requests per second
- ✓ Supports SSL/TLS termination

2 APPLICATION LOAD BALANCER (ALB)



Operates at Layer 7 (Application layer) of the OSI model. Routes HTTP and HTTPS traffic based on content such as path, host headers, and query strings.

KEY FEATURES

- ✓ Content-based routing (path, host, header)
- ✓ HTTP/2 and WebSocket support
- ✓ Integrated with AWS WAF
- ✓ Sticky sessions
- ✓ SSL/TLS termination
- ✓ Ideal for web applications and APIs

3 NETWORK LOAD BALANCER (NLB)



Operates at Layer 4 (Transport layer) of the OSI model. Handles TCP, UDP, and TLS traffic with ultra-low latency and high throughput.

KEY FEATURES

- ✓ Ultra-high performance
- ✓ Static IP support
- ✓ Handles millions of requests per second
- ✓ Preserves source IP
- ✓ Ideal for high-performance applications and microservices

4 AWS AUTO SCALING

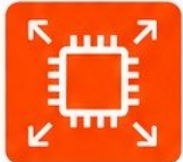


AWS Auto Scaling automatically adjusts the number of EC2 instances based on demand, ensuring optimal performance and cost efficiency.

KEY BENEFITS

- ✓ Automatically scales capacity to match demand
- ✓ Maintains application availability
- ✓ Helps control costs by scaling in when demand decreases
- ✓ Works with load balancers for high availability

5 EC2 AUTO SCALING



EC2 Auto Scaling manages the scaling of EC2 instances across one or more Availability Zones based on defined policies.

KEY FEATURES

- ✓ Launch template or configuration support
- ✓ Health checks and automatic replacements
- ✓ Multi-AZ support
- ✓ Seamless integration with ELB

6 SCALING POLICIES

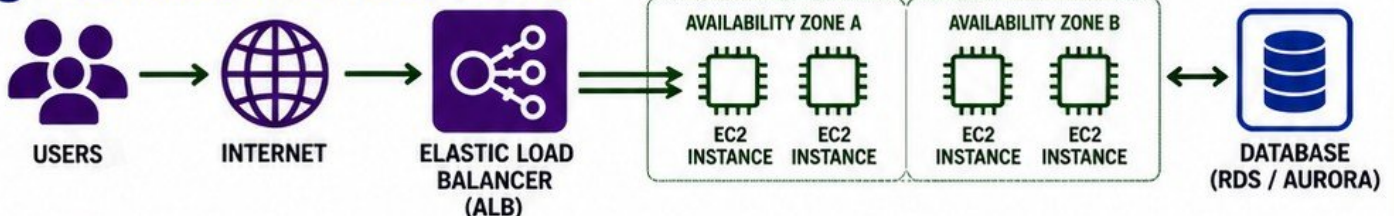


Scaling policies define when and how to scale your resources based on metrics, schedules, or custom conditions.

TYPES OF POLICIES

- ✓ Target Tracking Scaling (e.g., average CPU utilization)
- ✓ Step Scaling (scale based on metric thresholds)
- ✓ Scheduled Scaling (scale based on time and date)
- ✓ Predictive Scaling (uses machine learning to forecast demand)

7 HIGH AVAILABILITY ARCHITECTURE EXAMPLE



ELASTICITY

Scale out during high demand.
Scale in when demand drops.



HIGH AVAILABILITY

Spread across multiple
Availability Zones.



COST EFFICIENCY

Pay only for what you use.
Optimize resources automatically.

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10. SERVERLESS AND APPLICATION INTEGRATION



BUILD MODERN APPLICATIONS. INTEGRATE SERVICES. SCALE WITHOUT SERVERS.

1 AWS LAMBDA



Run code without provisioning or managing servers. Lambda automatically scales and you pay only for the compute time you use.

KEY FEATURES

- ✓ Serverless compute service
- ✓ Supports multiple languages
- ✓ Automatic scaling
- ✓ Pay per request and duration
- ✓ Integrated with many AWS services

2 AMAZON API GATEWAY



Create, publish, maintain, monitor, and secure APIs at any scale. API Gateway acts as the front door for applications.

KEY FEATURES

- ✓ REST, HTTP, and WebSocket APIs
- ✓ Request/response transformation
- ✓ Authentication and authorization
- ✓ Rate limiting and throttling
- ✓ Integrated with Lambda and other services

3 AMAZON SQS



Fully managed message queuing service that decouples and scales microservices, distributed systems, and serverless applications.

KEY FEATURES

- ✓ Decouples applications
- ✓ Reliable message delivery
- ✓ Standard and FIFO queues
- ✓ Message visibility timeout
- ✓ Dead-letter queues

4 AMAZON SNS



Fully managed pub/sub messaging service that delivers messages to multiple subscribers via multiple protocols.

KEY FEATURES

- ✓ Publish to multiple subscribers
- ✓ Email, SMS, HTTP, Lambda, SQS and more
- ✓ Fan-out messaging
- ✓ Message filtering
- ✓ High availability

5 AMAZON EVENTBRIDGE



Serverless event bus that enables you to connect applications using events. Built-in event sources and custom event patterns.

KEY FEATURES

- ✓ Real-time event routing
- ✓ Built-in and custom events
- ✓ Content-based filtering
- ✓ Event-driven orchestration
- ✓ Integrates with AWS services and SaaS apps

6 SERVERLESS COMPUTING

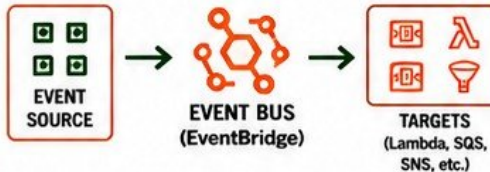


Serverless computing allows you to build and run applications without managing servers. The cloud provider handles provisioning, scaling, high availability, and maintenance.

BENEFITS

- ✓ No server management
- ✓ Automatic scaling
- ✓ High availability by default
- ✓ Reduced operational overhead
- ✓ Pay only for what you use
- ✓ Focus on code, not infrastructure

7 EVENT-DRIVEN ARCHITECTURE



Components communicate by producing and consuming events. This pattern enables loose coupling, scalability, and real-time responsiveness.

KEY ADVANTAGES

- ✓ Loosely coupled systems
- ✓ Scales automatically with demand
- ✓ Resilient and fault tolerant
- ✓ Real-time processing
- ✓ Easy to extend and integrate

8 COMMON SERVERLESS USE CASES



Web and mobile backends
APIs, user authentication, and real-time data processing.



Data processing and ETL
Process files, transform data, and load to data stores.



File and image processing
Resize images, convert formats, scan documents.



Chatbots and notifications
Send alerts, emails, SMS, and in-app messages.



IoT and real-time analytics
Collect device data, analyze, and trigger actions.

9 SERVERLESS INTEGRATION EXAMPLE



CLIENT
(MOBILE / WEB)



AMAZON API
GATEWAY



AWS LAMBDA
(BUSINESS LOGIC)



AMAZON SQS
(QUEUE)



AWS LAMBDA
(WORKER)



AMAZON SNS
(NOTIFICATIONS)



DATABASE
(DYNAMODB)

1 Client sends a request to API Gateway.

2 API Gateway forwards the request to Lambda.

3 Lambda processes the request and sends a message to SQS.

4 SQS decouples and stores the message securely.

5 Worker Lambda processes the message.

6 Results are stored in DynamoDB and/or notifications are sent via SNS.



SERVERLESS. EVENT-DRIVEN. SCALABLE BY DESIGN.
BUILD MORE. MANAGE LESS. DELIVER FASTER.

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11. CONTENT DELIVERY AND DNS



DELIVER CONTENT FASTER. IMPROVE PERFORMANCE. REACH USERS GLOBALLY.

1 AMAZON ROUTE 53



Amazon Route 53 is a highly available and scalable Domain Name System (DNS) web service that routes users to internet applications by translating domain names into IP addresses.

KEY FEATURES

- ✓ Domain registration and management
- ✓ DNS routing and traffic management
- ✓ Health checks to monitor application health
- ✓ High availability and global infrastructure
- ✓ Security with DNSSEC

2 DNS FUNDAMENTALS

- DNS translates domain names to IP addresses.
- Hierarchical and distributed system.
- Common Record Types:

A	Maps domain to IPv4 address
AAAA	Maps domain to IPv6 address
CNAME	Alias of another domain
MX	Mail exchange server
TXT	Text records for verification
NS	Name server records

3 AMAZON CLOUDFRONT



Amazon CloudFront is a fast content delivery network (CDN) service that securely delivers data, videos, websites, and APIs to users with low latency and high transfer speeds.

KEY FEATURES

- ✓ Global content delivery network (CDN)
- ✓ Secure content with SSL/TLS
- ✓ Integration with AWS services
- ✓ DDoS protection and security
- ✓ Detailed real-time monitoring

4 EDGE LOCATIONS



CloudFront has 600+ Edge Locations in 190+ countries.

Content is cached closer to users to reduce latency and improve performance.

5 CONTENT CACHING



- Content is stored in edge locations.
- Reduces origin load and bandwidth cost.
- Configurable cache behaviors and TTL.
- Improves response time for users.
- Supports compression and object caching.

6 GLOBAL CONTENT DELIVERY



Distribute content globally with high availability and reliability.

Ensure fast, consistent access to users anywhere in the world.

7 AWS GLOBAL ACCELERATOR OVERVIEW



AWS Global Accelerator improves the availability and performance of your applications for global users using the AWS global network.

KEY BENEFITS

- ✓ Uses AWS global edge network for static anycast IPs
- ✓ Improves performance by routing traffic via optimal path
- ✓ Enhances availability with health checks and failover
- ✓ Supports TCP, UDP, and HTTP/HTTPS traffic
- ✓ Works with AWS and on-premises applications

8 IMPROVING APPLICATION PERFORMANCE



- Use Route 53 for intelligent DNS routing and failover.
- Leverage CloudFront for caching and global delivery.
- Use Global Accelerator for consistent performance.
- Optimize caching, compression, and static content delivery.
- Monitor performance with CloudWatch and CloudFront metrics.

9 CONTENT DELIVERY FLOW EXAMPLE



USER



AMAZON ROUTE 53
(DNS RESOLUTION)



AWS GLOBAL
ACCELERATOR



AMAZON CLOUDFRONT
(EDGE LOCATION)



ORIGIN SERVER
(S3 / EC2 / ALB)

- ✓ Faster Load Times
- ✓ Lower Latency
- ✓ Higher Availability
- ✓ Better User Experience
- ✓ Global Reach



THE RIGHT DNS. GLOBAL DELIVERY. FASTER PERFORMANCE.
BUILD RELIABLE. DELIVER EVERYWHERE.

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12. MONITORING, MANAGEMENT, AND GOVERNANCE



MONITOR. MANAGE. SECURE. GOVERN YOUR AWS ENVIRONMENTS.

1 AMAZON CLOUDWATCH



Amazon CloudWatch monitors AWS resources and applications in real time.

KEY CAPABILITIES

- ✓ Metrics and dashboards
- ✓ Alarms and notifications
- ✓ Logs (CloudWatch Logs)
- ✓ Events and rules
- ✓ CloudWatch Agent for custom monitoring

2 AWS CLOUDTRAIL



AWS CloudTrail logs all API calls and events across your AWS account.

KEY CAPABILITIES

- ✓ Records API activity
- ✓ Tracks user and service actions
- ✓ Security and compliance auditing
- ✓ Event history and search
- ✓ Integration with CloudWatch Logs and S3

3 AWS CONFIG



AWS Config assesses, audits, and evaluates the configurations of your AWS resources.

KEY CAPABILITIES

- ✓ Resource inventory
- ✓ Configuration history
- ✓ Change tracking
- ✓ Compliance evaluation
- ✓ Custom rules and remediation

4 AWS SYSTEMS MANAGER



AWS Systems Manager helps you operate, manage, and secure your AWS resources at scale.

KEY CAPABILITIES

- ✓ Run Command
- ✓ Patch Manager
- ✓ Session Manager
- ✓ Parameter Store
- ✓ Automation for operations

5 AWS ORGANIZATIONS



AWS Organizations helps you centrally manage multiple AWS accounts.

KEY CAPABILITIES

- ✓ Create and manage accounts
- ✓ Organize accounts into OUs
- ✓ Consolidated billing
- ✓ Centralized policy management
- ✓ Enable security and governance at scale

6 SERVICE CONTROL POLICIES (SCPs)



SCPs allow you to define the maximum permissions for accounts within your organization.

KEY FEATURES

- ✓ Set guardrails for accounts
- ✓ Deny or restrict AWS services and actions
- ✓ Apply at root, OU, or account level
- ✓ Prevent unwanted or unauthorized actions

7 AWS CONTROL TOWER OVERVIEW



AWS Control Tower sets up and governs a secure, compliant, multi-account AWS environment quickly.

KEY FEATURES

- ✓ Automated account setup
- ✓ Baseline security guardrails
- ✓ Centralized logging and auditing
- ✓ Pre-built blueprints
- ✓ Continuous compliance

8 MONITORING VS AUDITING



MONITORING

Real-time visibility of system and application performance and health.

AUDITING

Recording and reviewing activities and configuration changes for security, compliance, and governance.

9 MONITORING AND GOVERNANCE WORKFLOW



MONITOR

Use CloudWatch to monitor metrics, logs, and events in real time.



RECORD

CloudTrail records all API activity across your environment.



EVALUATE

AWS Config evaluates resource configurations and compliance continuously.



MANAGE

Systems Manager automates operations, patching, and maintenance.



GOVERN

Organizations and SCPs enforce policies and guardrails across all accounts.



STANDARDIZE

Control Tower provides a secure, governed, and compliant landing zone at scale.

MONITORING (REAL-TIME OPERATIONS)

- Focuses on performance, availability, and health.
- Uses CloudWatch metrics, logs, and alarms.
- Detects issues and triggers alerts.
- Example: CPU utilization alarm, application latency.

VS

AUDITING (HISTORY & COMPLIANCE)

- Focuses on who did what, when, and where.
- Uses CloudTrail logs and Config history.
- Supports security reviews and compliance.
- Example: API call history, configuration changes.



**GOOD MONITORING KEEPS YOU INFORMED.
 GOOD GOVERNANCE KEEPS YOU PROTECTED.**

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13. AWS SECURITY SERVICES



SECURE YOUR WORKLOADS. PROTECT YOUR DATA. STRENGTHEN YOUR CLOUD.

1 AWS KMS



AWS Key Management Service makes it easy to create and control the encryption keys used to encrypt your data.

KEY FEATURES

- ✓ Create and manage encryption keys
- ✓ Encrypt and decrypt data
- ✓ Integrates with many AWS services
- ✓ Key rotation
- ✓ Fine-grained key policies

2 AWS SECRETS MANAGER



Store, retrieve, and rotate secrets such as database credentials, API keys, and passwords securely.

KEY FEATURES

- ✓ Secure secret storage
- ✓ Automatic secret rotation
- ✓ Fine-grained access control
- ✓ Integration with AWS services
- ✓ Audit and monitoring

3 AMAZON GUARDDUTY



Intelligent threat detection service that continuously monitors for malicious activity and unauthorized behavior.

KEY FEATURES

- ✓ ML-powered threat detection
- ✓ Detects compromised instances
- ✓ Detects malicious IPs
- ✓ S3 protection
- ✓ Low setup and maintenance

4 AWS SECURITY HUB



Provides a comprehensive view of your security posture by aggregating findings from AWS and partner services.

KEY FEATURES

- ✓ Centralized security findings
- ✓ Aggregates AWS and third-party tools
- ✓ Standardized security checks
- ✓ Custom insights
- ✓ Integration with AWS services

5 AMAZON INSPECTOR



Automated security assessment service that helps improve the security and compliance of your applications.

KEY FEATURES

- ✓ Vulnerability assessment
- ✓ CIS benchmark checks
- ✓ Network reachability analysis
- ✓ Agentless or agent-based
- ✓ Detailed findings and reports

6 AWS SHIELD



Managed DDoS protection service that safeguards your applications running on AWS.

KEY FEATURES

- ✓ Standard and Advanced tiers
- ✓ Always-on DDoS protection
- ✓ Automatic traffic monitoring
- ✓ Cost protection (Advanced)
- ✓ Seamless with AWS services

7 AWS WAF



Web Application Firewall that helps protect your web applications from common web exploits.

KEY FEATURES

- ✓ Filter HTTP/S traffic
- ✓ OWASP Top 10 protection
- ✓ IP reputation and rate limiting
- ✓ Custom rules
- ✓ Integration with CloudFront, ALB, API Gateway

8 AWS ARTIFACT



On-demand access to AWS compliance reports and agreements to support your audits.

KEY FEATURES

- ✓ Access compliance reports
- ✓ Access security agreements
- ✓ ISO, SOC, PCI, HIPAA, and more
- ✓ Audit readiness
- ✓ Region-specific documents

9 ENCRYPTION FUNDAMENTALS



Encryption protects your data by converting it into an unreadable format. Only authorized users with the correct key can decrypt and access the data.



10 SECURITY IN THE AWS SHARED RESPONSIBILITY MODEL



CUSTOMER RESPONSIBILITY

- Data classification and encryption
- Identity and access management
- Operating systems and applications
- Network security (e.g., security groups)
- Patching and configuration of instances
- Monitoring and logging of resources

VS



AWS RESPONSIBILITY

- Physical data centers
- Hardware, networking, and virtualization
- Infrastructure availability
- Managed services availability
- Security of the cloud
- Compliance and certifications



SECURITY IS EVERYONE'S RESPONSIBILITY.
USE AWS SECURITY SERVICES. STAY PROTECTED. STAY COMPLIANT.

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14. RELIABILITY AND DISASTER RECOVERY

BUILD RESILIENT SYSTEMS. PREPARE FOR THE UNEXPECTED. RECOVER WITH CONFIDENCE.

1 HIGH AVAILABILITY



Ensure your systems are accessible and operational for users when needed.

Minimize downtime by designing for redundancy.

2 FAULT TOLERANCE



Continue operating even if one or more components fail.

No single point of failure.

3 RESILIENCE



Automatically recover from failures and adapt to changes.

Maintain service under all conditions.

4 BACKUP AND RESTORE



Regular backups protect your data from loss.

Restore quickly when needed.

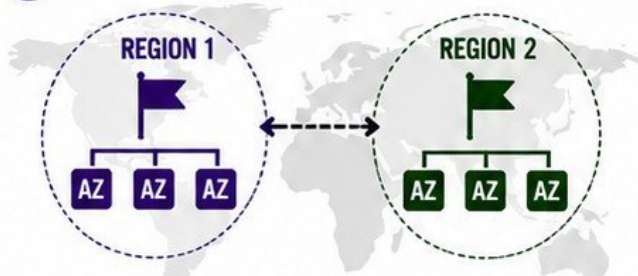
5 MULTI-AZ ARCHITECTURE



Deploy resources across multiple Availability Zones in a Region.

Protects against AZ failures.

6 MULTI-REGION CONCEPTS



Deploy across multiple Regions for greater availability, latency optimization, and disaster recovery.

7 RECOVERY TIME OBJECTIVE (RTO)



The maximum acceptable downtime after a disaster before systems must be restored.

FOCUS:
HOW FAST YOU
MUST RECOVER

8 RECOVERY POINT OBJECTIVE (RPO)



The maximum acceptable amount of data loss measured in time.

FOCUS:
HOW MUCH DATA
YOU CAN LOSE

9 DISASTER RECOVERY STRATEGIES

BACKUP AND RESTORE (BR)



Back up data regularly and restore on new infrastructure.

LOW COST
HIGH RECOVERY TIME

PILOT LIGHT



Keep core infrastructure running; scale out resources during a disaster.

BALANCED COST
MODERATE RTO

WARM STANDBY



Maintain a scaled-down copy of the environment ready to handle production traffic.

HIGHER COST
LOWER RTO

MULTI-SITE ACTIVE-ACTIVE



Traffic is served from multiple sites simultaneously.

HIGHEST COST
LOWEST RTO

MULTI-SITE ACTIVE-PASSIVE



Primary site handles traffic; secondary site stands by for failover.

HIGH COST
LOW RTO



PLAN FOR FAILURE. DESIGN FOR RECOVERY.
BUILD RELIABLE SYSTEMS. KEEP YOUR BUSINESS RUNNING.

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15. AWS WELL-ARCHITECTED FRAMEWORK



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 BUILD SECURE, HIGH-PERFORMING, RESILIENT, EFFICIENT, AND SUSTAINABLE SYSTEMS.
 MAKE INFORMED ARCHITECTURE DECISIONS.

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1 OPERATIONAL EXCELLENCE



- ✓ Run and monitor systems effectively.
- ✓ Improve processes and procedures.
- ✓ Respond to events and change.
- ✓ Learn from failures and successes.

2 SECURITY



- ✓ Protect information, systems, and assets.
- ✓ Implement least privilege access.
- ✓ Detect security events.
- ✓ Respond and recover from incidents.

3 RELIABILITY



- ✓ Ensure workload performs correctly.
- ✓ Automate to recover from failure.
- ✓ Scale to meet demand.
- ✓ Test recovery procedures regularly.

4 PERFORMANCE EFFICIENCY



- ✓ Use resources efficiently.
- ✓ Match performance to requirements.
- ✓ Optimize resource selection.
- ✓ Monitor and improve performance.

5 COST OPTIMIZATION



- ✓ Understand and manage costs.
- ✓ Avoid unnecessary costs.
- ✓ Scale cost-efficiently.
- ✓ Continuously optimize spending.

6 SUSTAINABILITY



- ✓ Minimize environmental impact.
- ✓ Improve energy efficiency.
- ✓ Reduce waste.
- ✓ Design for long-term sustainability.

7 WELL-ARCHITECTED REVIEWS



- ✓ Evaluate your architecture against the 6 pillars.
- ✓ Identify risks and improvement opportunities.
- ✓ Get expert guidance and actionable feedback.
- ✓ Iterate and improve continuously.

8 BASIC ARCHITECTURE DECISIONS



- ✓ Define requirements and constraints.
- ✓ Explore design options.
- ✓ Evaluate trade-offs using the 6 pillars.
- ✓ Choose the best option for your workload.

THE 6 PILLARS WORK TOGETHER



WELL-ARCHITECTED SYSTEMS ARE **SAFE, RELIABLE, EFFICIENT, AND SUSTAINABLE.**
REVIEW. IMPROVE. EVOLVE.

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16. AWS PRICING AND COST MANAGEMENT

PAY ONLY FOR WHAT YOU USE. OPTIMIZE CONTINUOUSLY. MAXIMIZE VALUE.

1 PAY-AS-YOU-GO PRICING



No upfront costs. Pay only for the resources you use. No long-term commitments.

2 ON-DEMAND PRICING



Ideal for short-term workloads and unpredictable usage. No commitment required.

3 SAVINGS PLANS



Commit to a consistent usage (1 or 3 years) for significant savings on compute usage.

4 RESERVED INSTANCES



Reserve instances for 1 or 3 years for up to 72% discount compared to On-Demand.

5 SPOT INSTANCES



Use spare AWS capacity at up to 90% discount. Ideal for flexible, fault-tolerant workloads.

6 AWS PRICING CALCULATOR



Estimate costs for your AWS services and architectures before you build.

calculator.aws

7 AWS COST EXPLORER



Analyze and visualize your costs over time to identify trends and anomalies.

aws.amazon.com/cost-explorer

8 AWS BUDGETS



Set custom budgets and alerts to stay on track and avoid surprises.

aws.amazon.com/budgets

9 COST ALLOCATION TAGS



Tag your resources to track costs by department, team, project, or environment.

aws.amazon.com/resource-tags



COST ALLOCATION TAGS

Use tags to categorize and allocate costs accurately across teams, projects, and environments for better visibility and accountability.

TAG KEY	EXAMPLE VALUE	BENEFIT
Department	Engineering	Track costs by department
Project	Website Migration	Track costs by project
Environment	Production	Track costs by environment
Owner	DevOps Team	Assign cost ownership
Cost Center	IT-Operations	Map to your business structure

BEST PRACTICES FOR COST MANAGEMENT



Monitor and analyze costs regularly.



Right-size resources to match demand.



Use Savings Plans and Reserved Instances.



Set budgets and alerts to control costs.

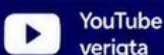


Tag all resources consistently.



UNDERSTAND YOUR COSTS. CONTROL YOUR SPEND.
OPTIMIZE CONTINUOUSLY. DELIVER MAXIMUM VALUE.

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17. AWS SUPPORT AND CLOUD ECONOMICS

GET THE RIGHT SUPPORT. UNDERSTAND THE VALUE OF THE CLOUD.

1 AWS SUPPORT PLANS

BASIC



Included at no extra cost

- ✓ Account and billing support
- ✓ Access to AWS Trusted Advisor core checks
- ✓ Access to AWS re:Post
- ✓ Service health checks

DEVELOPER



For developers experimenting in AWS

- ✓ Business hours support
- ✓ 1 primary contact
- ✓ Shorter response times
- ✓ Trusted Advisor full checks

BUSINESS



For workloads in production

- ✓ 24/7 support
- ✓ Unlimited contacts
- ✓ Faster response times
- ✓ Architectural guidance

ENTERPRISE



For mission-critical workloads

- ✓ 24/7 support with dedicated team
- ✓ Technical Account Manager
- ✓ Proactive guidance
- ✓ Best response times

2 AWS TRUSTED ADVISOR



Real-time guidance to help you provision your resources following AWS best practices.

- ✓ Cost optimization
- ✓ Performance
- ✓ Security
- ✓ Fault tolerance
- ✓ Service limits

3 AWS HEALTH



Personalized view of AWS service health and events that may impact you.

- ✓ Service health dashboard
- ✓ Scheduled changes
- ✓ Event notifications
- ✓ Historical events

4 AWS RE:POST



Ask questions. Get answers. Learn from experts and the AWS community.

- ✓ Community Q&A
- ✓ Knowledge sharing
- ✓ Peer-to-peer support
- ✓ AWS expert participation

5 TOTAL COST OF OWNERSHIP (TCO)



The full cost of building, running, and managing solutions.

- ✓ Direct costs
- ✓ Indirect costs
- ✓ Operational overhead
- ✓ Risk and downtime

6 CLOUD ECONOMICS

CAPEX vs OPEX



CAPEX

Upfront investment. You own the asset. Higher risk.

OPEX

Pay only for what you use. Lower risk and flexible.

ECONOMIES OF SCALE



AWS spreads infrastructure costs across millions of customers to lower prices.

- ✓ Lower unit costs
- ✓ Continuous price drops
- ✓ More value for less

CLOUD FINANCIAL BENEFITS



- ✓ No upfront hardware costs
- ✓ Pay-as-you-go
- ✓ Reduce operational costs
- ✓ Scale up or down quickly
- ✓ Focus on innovation
- ✓ Improve cash flow

WHY IT MATTERS



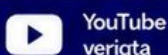
Better support + smart economics = lower costs, higher reliability, and more business value.

- ✓ Optimize spend
- ✓ Increase agility
- ✓ Drive better outcomes


**THE RIGHT SUPPORT KEEPS YOU RUNNING.
 THE RIGHT ECONOMICS KEEPS YOU GROWING.**

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18. AWS MIGRATION AND CLOUD ADOPTION



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MOVE WORKLOADS. MODERNIZE SYSTEMS. TRANSFORM YOUR BUSINESS.

1 AWS CLOUD ADOPTION FRAMEWORK (CAF)



A proven framework to align people, processes, and technology for successful cloud adoption.

- ✓ Business perspective
- ✓ People perspective
- ✓ Governance perspective
- ✓ Platform perspective
- ✓ Security perspective
- ✓ Operations perspective

2 MIGRATION FUNDAMENTALS



Understand the key concepts, benefits, and best practices of migrating to AWS.

- ✓ Why migrate to the cloud
- ✓ Business and technical drivers
- ✓ Lift and shift vs modernization
- ✓ Total cost and TCO benefits
- ✓ Risk and dependency analysis

3 AWS MIGRATION HUB



Simplify migration tracking and management at scale.

- ✓ Centralized migration dashboard
- ✓ Track migration progress
- ✓ Discover and assess servers
- ✓ Group, prioritize, and plan
- ✓ Integrate tools and services

4 AWS APPLICATION MIGRATION SERVICE



Lift and shift applications with minimal changes and reduced downtime.

- ✓ Continuous replication
- ✓ Automated orchestration
- ✓ Minimal downtime
- ✓ Cutover with confidence
- ✓ Physical, virtual, or cloud to AWS

5 AWS DATABASE MIGRATION SERVICE



Migrate databases quickly and securely with minimal downtime.

- ✓ Support for many database engines
- ✓ Continuous data replication
- ✓ Automated schema conversion
- ✓ Low downtime migrations
- ✓ Ongoing replication for cutover

6 AWS SNOW FAMILY



SNOWCONE



SNOWBALL



SNOWMOBILE

Move large amounts of data in or out of AWS securely and efficiently.

- ✓ Snowcone – Small data transfer
- ✓ Snowball – Petabyte-scale transfer
- ✓ Snowmobile – Exabyte-scale transfer
- ✓ Secure transport and tracking
- ✓ Ideal for offline data transfer

7 MIGRATION STRATEGIES

REHOST (LIFT AND SHIFT)



Move applications to AWS without changing architecture. Fastest way to get started.

REPLATFORM (LIFT, TINKER, AND SHIFT)



Make some optimizations to improve performance and efficiency.

REFACTOR (REARCHITECT)



Redesign applications to fully leverage cloud-native services.

REPURCHASE (REPLACE)



Replace with SaaS or managed services where possible.

RETAIN



Keep some workloads on-premises or other environments.

RETIRE



Eliminate unused applications and reduce costs.

8 BASIC CLOUD TRANSFORMATION CONCEPTS



- ✓ Align business goals with cloud strategy
- ✓ Embrace agility, innovation, and automation
- ✓ Modernize applications and data platforms
- ✓ Leverage managed and serverless services
- ✓ Use data-driven decision making
- ✓ Build scalable, secure, and resilient systems
- ✓ Continuously optimize and improve
- ✓ Drive business value and better customer outcomes

CLOUD TRANSFORMATION



People + Process + Technology = Value



MIGRATE WITH CONFIDENCE. MODERNIZE WITH PURPOSE.
TRANSFORM FOR THE FUTURE.

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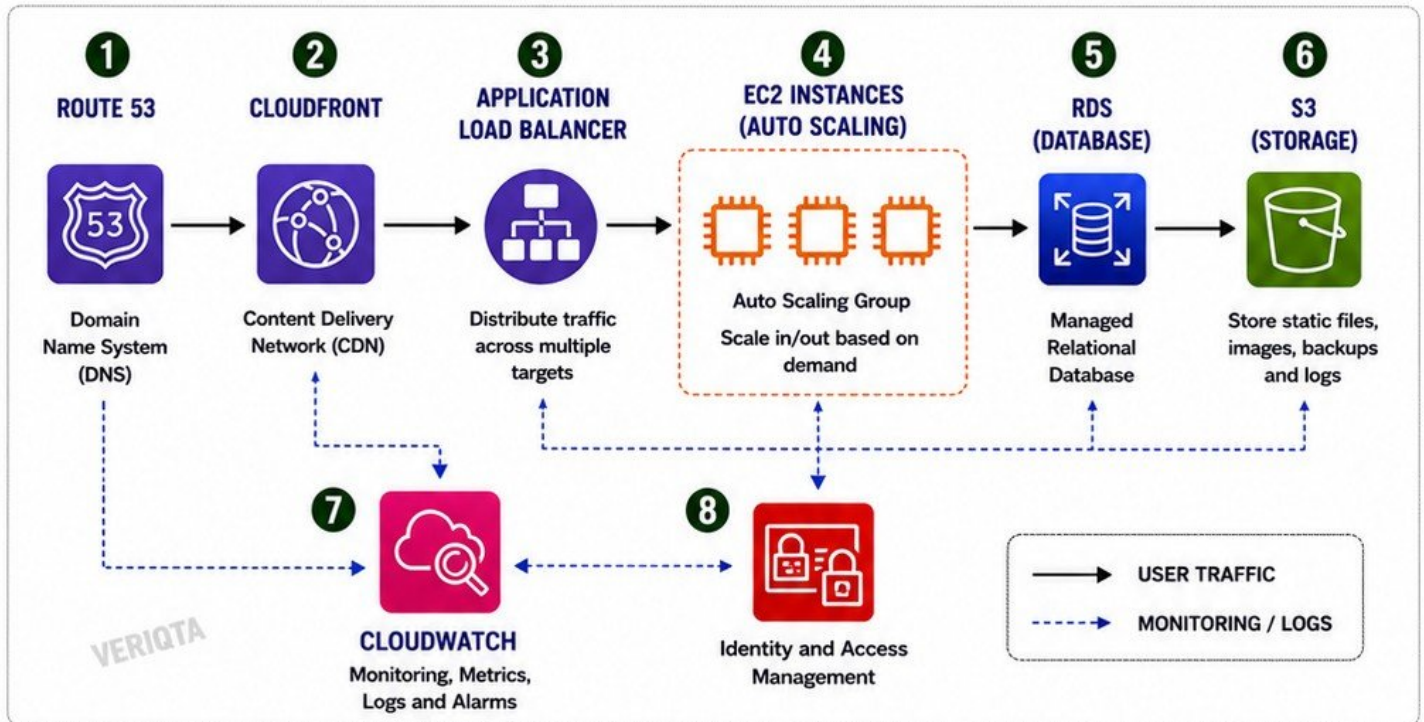
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19. REAL-WORLD AWS ARCHITECTURE LAB

BUILD A SIMPLE WEB APPLICATION ARCHITECTURE



SECURITY CONSIDERATIONS

- ✓ Use IAM roles and least privilege access
- ✓ Enable HTTPS with SSL/TLS (ACM)
- ✓ Use AWS WAF to protect web traffic
- ✓ Keep EC2 instances in private subnets
- ✓ Use security groups and NACLs
- ✓ Encrypt data at rest (RDS, S3) and in transit
- ✓ Enable CloudTrail for audit logging



AVAILABILITY CONSIDERATIONS

- ✓ Use Route 53 for DNS failover
- ✓ Use CloudFront for high availability
- ✓ ALB across multiple Availability Zones
- ✓ Auto Scaling ensures capacity and uptime
- ✓ RDS Multi-AZ for database redundancy
- ✓ S3 is highly durable and available by design
- ✓ CloudWatch alarms for proactive response



COST CONSIDERATIONS

- ✓ Use CloudFront to reduce data transfer costs
- ✓ Enable S3 Lifecycle policies for old data
- ✓ Use right-sized EC2 instances
- ✓ Auto Scaling to match demand
- ✓ Use RDS Reserved Instances
- ✓ Monitor usage and set AWS Budgets
- ✓ Use Cost Explorer to analyze costs



ARCHITECTURE SUMMARY

This architecture delivers a highly available, scalable, and secure web application on AWS. Users access the app through Route 53 and CloudFront, traffic is distributed via ALB to EC2 instances. Data is stored in RDS and static content in S3. CloudWatch monitors the environment and IAM manages access securely.



HIGH
AVAILABILITY



STRONG
SECURITY



SCALABLE
ARCHITECTURE



COST
EFFECTIVE



DESIGN SMART. BUILD RELIABLE. RUN SECURE. OPTIMIZE COSTS.

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20. FINAL REVIEW AND EXAM PREPARATION



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MASTER THE BASICS. PASS WITH CONFIDENCE.

1

CORE AWS SERVICES RECAP

- Compute – EC2, Lambda, ECS
- Storage – S3, EBS, EFS
- Database – RDS, DynamoDB
- Networking – VPC, Route 53, CloudFront
- Security – IAM, KMS, Security Hub, WAF
- Management – CloudWatch, CloudTrail, Config
- Migration – Migration Hub, DMS, Snow Family

2

SERVICE COMPARISON CHEAT SHEET

S3	Object storage Scalable, durable, high availability
EBS vs EFS	EBS – Block storage EFS – Shared file storage
RDS vs DynamoDB	RDS – Relational DB DynamoDB – NoSQL DB
CloudFront vs ALB	CloudFront – Global CDN ALB – Load balancer for applications
EC2 vs Lambda	EC2 – Manage servers Lambda – Serverless compute

3

SECURITY RESPONSIBILITY CHECKLIST

- ☒ Understand the AWS Shared Responsibility Model
- ☒ Use IAM roles and least privilege
- ☒ Enable MFA for root user
- ☒ Encrypt data at rest and in transit
- ☒ Use Security Groups and NACLs correctly
- ☒ Enable CloudTrail and monitor logs
- ☒ Use AWS WAF for web protection
- ☒ Keep systems and applications updated
- ☒ Back up your data

4

PRICING MODEL CHEAT SHEET

- Pay-as-you-go
Pay for only what you use. No upfront commitment.
- On-Demand
Flexible. Highest cost.
- Savings Plans
Commit to usage for 1 or 3 years. Lower cost than On-Demand.
- Reserved Instances
Commit to specific instance for 1 or 3 years. Biggest savings.
- Spot Instances
Use spare AWS capacity. Up to 90% discount.

5

ARCHITECTURE DECISION QUESTIONS

- ? Which service should be used to store static website content?
- ? Which database is best for a highly scalable NoSQL application?
- ? How would you distribute traffic across multiple AZs?
- ? Which service provides serverless compute?
- ? How do you secure data in transit and at rest?
- ? Which service would you use for global content delivery?
- ? How would you design for high availability and fault tolerance?

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COMMON EXAM TRAPS

- ✗ Choosing the most complex service instead of the simplest.
- ✗ Ignoring cost-effective options.
- ✗ Confusing security responsibilities (AWS vs Customer).
- ✗ Not understanding service limits and quotas.
- ✗ Overlooking managed services that reduce operational effort.
- ✗ Assuming all services are available in all Regions.

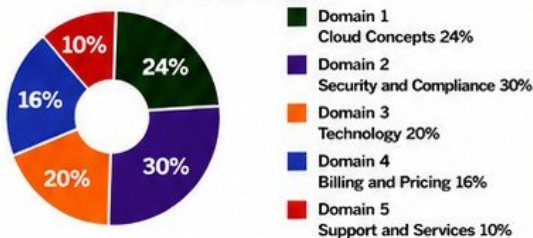
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AWS TERMINOLOGY REVIEW

Account – Your isolated AWS environment
 Region – A geographical area with multiple AZs
 AZ (Availability Zone) – Isolated location within a Region
 VPC – Virtual Private Cloud
 IAM – Identity and Access Management
 SLA – Service Level Agreement
 RTO – Recovery Time Objective
 RPO – Recovery Point Objective
 Elasticity – Automatically scale up or down
 Durability – Data is safe from loss

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CLOUD PRACTITIONER EXAM DOMAINS



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FINAL PRACTICE SCENARIOS

- ☒ Design a highly available web application.
- ☒ Choose the best storage solution for different use cases.
- ☒ Identify cost optimization strategies.
- ☒ Implement security best practices.
- ☒ Plan a migration to AWS.

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NEXT LEARNING PATH



Deepen Your Knowledge
 Explore AWS whitepapers, documentation, and hands-on labs.



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